OULAD Proposal

Objective:

- Test how well a student's background and their click activity predict their performance in their respective courses.

Data:

- Student information:
  - Demographics, registration dates, course length
- Assessments:
  - Due dates, weights, submission dates, grades, final result
- Clickstream:
  - ~10.6 millions rows of daily clicks per page with activity. The goal is to transform the clicking data into one row per student-course before joining.

Unit of Analysis: One student in one course run. 3,538 students took more than one course, so they have more than one row.

Outcome: Pass (Pass or Distinction), Fail or Withdrawn.

Predictors: Background known at registration, plus activity and assessment features from before day 28.

Cutoff:

- **Initial settings:** Day 0 is the course start, and only data dated on and before day 28 is used. Students who withdrew before day 28 are left out because their outcome is already known. This leaves 27,535 rows: 15,379 Pass, 7,037 Fail and 5,119 Withdrawn.
- **Later settings:** model training on different cutoff dates (example: 2, 4, 6 weeks), then compare which one performs better.

Features

Group	Features	Reason
Background	Gender, region, age, education, IMD band, disability, previous attempts, credits, registration date, course and term	Known at the start; needed for subgroup checks
Click volume	Total clicks, active days, clicks per week, pages visited	Overall engagement

Click timing	Clicks before day 0, first and last active day, trend across weeks	Separates early starters from students fading out
Click type	Clicks by activity type (8 main types plus other)	Forum and quiz use may matter more than page views
Relative clicks	Percentile among classmates in the same course	Click volume differs by course
Assessments	Number due, submitted and missed; mean and weighted score; days early or late	Missed work and low early scores are warning signs

What data to leave out:

Item	**Reason**
date_unregistration	Reveals who withdrew. Used only to choose who is included.
Students who withdrew before day 28 (5,040) or registered after it (18)	Outcome already known, or not enrolled yet
Clicks and scores from day 28 on	Not known at the cutoff
Exam results	Taken at the end of the course
vle week_from and week_to	82% missing
id_student	Identifier only

Methods

Python (pandas, scikit-learn) in Jupyter notebooks. Or R in rmd.

Required element	**Our plan**
Exploratory analysis	Class balance; clicks and scores by outcome; differences between courses.
Missing-data handling	Missing IMD band becomes "Unknown". Other gaps are filled inside training folds.

Validation without leakage	Only data before the cutoff. Folds grouped by student and stratified by outcome; nested cross-validation; preprocessing fit inside training folds.
Regularized model	Elastic net logistic regression. Click features overlap heavily, and the penalty keeps the model stable. Coefficients are easy to explain.
Tree-based model	Gradient boosting (HistGradientBoostingClassifier). Usually accurate on tabular data, handles missing values directly and picks up effects that differ by course.
Baseline	Predict Pass; logistic regression on background only.
Discrimination	AUROC per class, balanced accuracy and confusion matrix, also by course and subgroup.
Calibration	Reliability plots, log loss and Brier score; recalibrate inside training folds if needed.
Cutoff comparison	Same pipeline at 2, 4 and 6 weeks.

Implemented wide-table transformations (day 28 inclusive)

Area	**Transformation completed**
Leakage controls	Removed date_unregistration, exam fields, assessment IDs and all scores (grade-release times unavailable).
Submission window	Replaced full-course submission fields with current-attempt submission dates/flags through day 28; retained pre-course submissions; excluded banked credits.
Missing IMD band	1,111 missing values → Unknown category.
Missing registration date	45 missing values → −54 days (eligible 2013 median), with a missingness indicator; refit imputation within training folds.
Structural assessment gaps	Retained planned non-exam dates/weights; assessments not offered → 0 with an offered indicator.
No early submission	Submission day → 0 with submitted flag = 0; distinguishes absence from a genuine day-zero submission.
Cohort and model roles	Separate eligibility file flags 27,422 eligible enrolments; final_result is the outcome; student ID, split and cohort flags are not predictors.
Output and checks	Original backed up; all 32,593 rows retained; 105 → 80 columns; no missing cells; keys and outcome labels unchanged.

Proposed division of remaining work

Completed contributions: Ahmed—table widening/concatenation and initial click-feature planning; Daniel—remaining data transformations, missingness handling and leakage controls.

Workstream	Lead	Deliverable
Analysis setup	Both	Agree outcome coding, cutoff boundary, cohort and shared validation splits; reserve the final test set.
Clickstream features	Ahmed	Implement cutoff-limited clicks, active days, recency, weekly trends and activity-type summaries; merge and check enrolment keys.
Exploratory analysis	Ahmed	Summarize class balance, course differences and engagement patterns; flag outliers and redundant click features.
Baseline and regularized model	Daniel	Build fold-specific preprocessing, a majority-class baseline and elastic-net logistic regression; tune within training folds.
Tree-based model	Ahmed	Fit and tune gradient boosting using the same features, folds and outcome as the regularized model.
Evaluation and subgroup checks	Daniel	Compare discrimination, confusion matrices and calibration; report performance by course and demographic subgroup with uncertainty.
Robustness checks	Ahmed / Daniel	Ahmed: temporal and cross-course tests. Daniel: cutoff and cohort sensitivity analyses. Apply one shared evaluation protocol.
Report and presentation	Both	Ahmed: data/click features and boosting. Daniel: preprocessing, validation and interpretation. Both: discussion, slides and final review.

Both review each other’s code and interpretation before final submission.